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Implementing learning design with learning analytics to scaffold the regulation of collaboration in primary education

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Abstract

The regulation of learning, including self-regulated learning (SRL) and socially shared regulation of learning (SSRL), is a key skill that should already be practiced in primary education. The shift toward student-centered, collaborative, and open-ended processes, such as phenomenon-based learning, increases the need for pupils to take charge of their own learning. However, due to the varying abilities of primary school pupils to regulate their learning, such processes can be challenging and require substantial scaffolding. To address this issue, this study explores how a learning design that indirectly scaffolds the regulation of collaboration can support regulation during the challenging phases of a collaborative blended learning process in primary school. Additionally, the potential of learning analytics (LA) to further support pupils' processes is discussed. A specific study module on sustainable development was designed and implemented in a learning management system (LMS). Classroom observations and LMS log data were collected during the implementation with fifth- and sixth-grade pupils. The observation data show that the most challenging phases of the group work occurred at the beginning of the group process and again during the finalization of the group project. The log data further indicate that pupils used the LMS as a regulation support at the beginning of the process. Overall, the results highlight the need for more functional tools that help groups to metacognitively monitor and summarize their learning. In addition, LA could provide meaningful visualizations for teachers to further develop learning designs during learning processes.

Keywords: Self-regulated learning, Learning design, Collaborative learning, Learning analytics, Primary education, Scaffolding

Introduction

In today's landscape of continuous and digital learning, alongside the growing influence of artificial intelligence (AI) in education, self-regulated learning (SRL) is widely recognized as a key factor in educational success, starting from primary school (Dignath & Büttner, 2008; Panadero, 2017). In recent decades, student-centered and digitally enhanced learning methods—where students work collaboratively in groups—have transformed educational practices in primary classrooms. This shift is particularly evident in Finnish basic education, where the national curriculum strongly promotes collaborative learning through a phenomenon-based learning approach (Finnish National Board of Education, 2014). While these pedagogical developments provide learners with greater autonomy and opportunities to plan and make decisions about their learning, they also present certain challenges. For effective collaboration, students must manage their own learning and take responsibility for their peers' contributions (Järvelä et al., 2015).

In primary education, supporting students' self-regulation during group work is especially critical, as group dynamics can be particularly challenging for younger pupils. However, despite the recognized importance of SRL and collaborative regulation, research on how these skills can be scaffolded in primary schools remains limited (Dignath et al., 2008; Prasse et al., 2024). Given the heterogeneous nature of primary pupils' ability to regulate their learning (Kontturi, 2016; Liu et al., 2024), as well as the need for support tailored to learners' age and skill levels, further research into younger age cohorts is imperative. This is particularly significant as primary pupils are at a crucial stage for the development of their self-regulation strategies.

Previous research has shown that certain elements of the learning environment, especially learning design, significantly influence opportunities for SRL to emerge (Dignath & Veenman, 2021; Perry & VandeKamp, 2000; Vosniadou et al., 2024). Learning design, understood as the sequencing of educational materials and tasks, is critical not only for enhancing students' SRL (Paris & Paris, 2001) and the development of SRL strategies (Fan et al., 2021) but also for facilitating collaboration within groups (Järvelä et al., 2015). The implementation of structured scaffolding is therefore essential for assisting primary pupils in effectively navigating and achieving success in collaborative and open-ended learning processes. A learning design that incorporates structured, indirect scaffolding for learning regulation has been shown to effectively facilitate primary pupils' SRL (cf. Dignath & Veenman, 2021).

Learning analytics (LA) and artificial intelligence (AI) offer further potential by providing temporal, data-driven information and feedback to scaffold pupils' learning. In particular, LA can provide feedback on how learning design is enacted throughout the learning process (Ifenthaler et al., 2018; Persico & Pozzi, 2015). However, important questions remain regarding the design and use of LA in collaborative learning processes, including what data should be collected at different stages of learning and how such data should be effectively visualized (Lämsä et al., 2021). These challenges become even more complex in blended and collaborative learning environments where pupils learn and collaborate in class while using technological devices (see, e.g., Bonk et al., 2006; Mikulecky, 2019), as is common in primary education. Advances in LA research, particularly across diverse pedagogical contexts, provide an important foundation for designing more intelligent systems that clarify the roles of technology, learners, and teachers within different learning processes (Molenaar, 2022).

Therefore, combining learning designs that scaffold regulation with LA-based information about temporal learning sequences may help teachers and pupils navigate the more challenging phases of collaborative processes. Nevertheless, empirical evidence remains scarce regarding how primary school groups collaboratively regulate their learning during these processes and how learning management systems (LMS) are used to support this regulation. This empirical study therefore aims to examine how an

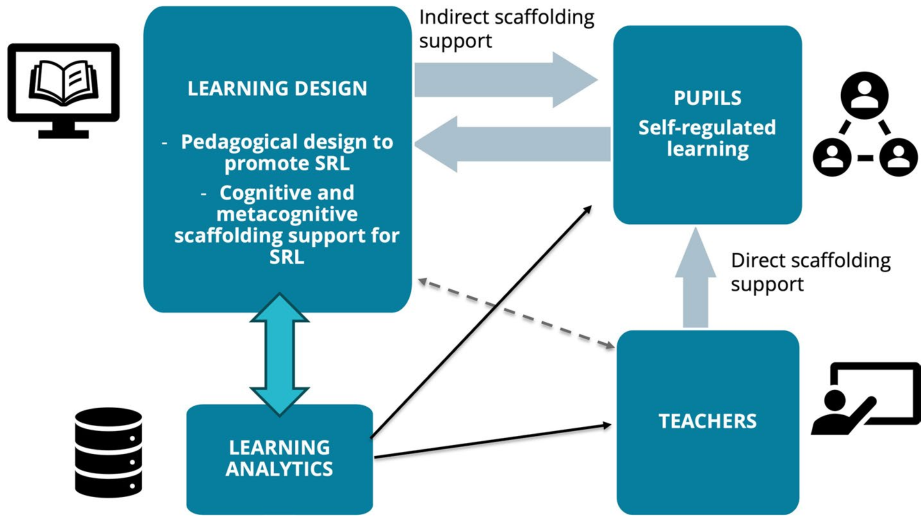

Fig. 1 Central concepts and elements for a learning design supporting the regulation of collaboration in classrooms in this study

LMS-based learning design can indirectly scaffold primary pupils' regulation of collaboration during challenging phases of collaborative, open-ended learning process. Additionally, the study explores how LA visualizations may further support these processes.

Theoretical background

In the following theoretical sections, we begin by outlining the theoretical background for the regulation of learning in collaborative processes. We then look at the pedagogical design principles used to promote the regulation of learning (see Fig. 1). Lastly, we discuss the specific supportive techniques (i.e. indirect scaffolding) that can help pupils regulate their collaboration. The methods section subsequently describes how these concepts were implemented in the study module.

Regulation of collaboration

Collaborative learning in primary schools involves pupils working together to complete assignments, learn new concepts, solve problems, and develop their collaboration skills. It emphasizes peer interaction and cooperation and is thought to lead to enhanced learning outcomes, the development of social skills (Slavin, 2014), increased motivation, and greater engagement (Gillies & Boyle, 2010). However, research shows that collaborative processes in primary schools do not always produce the intended outcomes. Common challenges include, for example, uneven participation and free-riding (cf. Hall & Buzwell, 2013), poor communication (Naykki et al., 2021), and lack of motivation (Järvenoja et al., 2019).

One explanation for these challenges is that pupils may lack sufficient skills to regulate their learning. In collaborative processes, pupils need to regulate their own learning and interactions (SRL) (Panadero, 2017; Zimmerman, 2002) and also support the regulation of the socially shared learning process (SSRL) (Panadero & Järvelä, 2015). SRL refers to the process by which individual students systematically activate and sustain their thoughts, feelings, motivations, and actions to attain their objectives (Schunk & Greene, 2018). In self-regulatory processes, students are active and responsible participants in their own learning and performance (Zimmerman, 2002). Zimmerman's cyclical model describes SRL as consisting of three phases: the forethought phase, involving goal setting and task analysis; the performance phase, during which learners apply regulatory strategies and monitor their progress; and the self-reflection phase, in which learners evaluate outcomes and processes (Zimmerman, 2002).

In contrast, SSRL describes regulation occurring collectively during group work, as group members jointly plan, monitor, and reflect on their learning (Panadero & Järvelä, 2015). While collaborating, groups establish shared goals and strategies and engage in shared metacognitive monitoring to achieve a common outcome (Hadwin et al., 2018; Panadero & Järvelä, 2015). During such processes, SRL and SSRL are intertwined, as individuals regulate their own learning while simultaneously participating in shared regulatory processes to reach collective goals (Hadwin et al., 2018). SSRL plays a critical role in successful collaborative learning, as it is associated with higher group performance, improved quality of collaboration, and enhanced individual learning outcomes (Järvelä et al., 2016). Consequently, when pupils possess heterogeneous regulatory skills, collaborative learning may become more challenging.

Although the regulation of learning is central to facilitating effective learning and achieving improved educational outcomes, learners typically do not engage in self-regulated learning spontaneously (Azevedo, 2005). This challenge is particularly evident in digital learning environments, which often provide open-ended navigation and therefore demand greater regulatory skills from learners (Azevedo, 2005). These conditions highlight the importance of effective learning design that leverages digital tools to scaffold pupils' SRL and SSRL.

Phenomenon-based learning design to promote the regulation of learning

Phenomenon-based learning, rooted in the Finnish education system, emphasizes holistic and inquiry-driven learning processes in which pupils explore real-world phenomena through collaborative engagement with their peers (Symeonidis & Schwarz, 2016). As these processes are typically highly student-centered, they are considered to support pupils in taking charge of their own learning, ultimately leading to improved self-regulation in educational contexts. However, since primary pupils demonstrate varying levels of self-regulation skills, phenomenon-based learning processes must be carefully designed to support SRL effectively. A properly planned learning design, which learners can rely on throughout the learning process, is recognized as a crucial link between learning sciences and environments that enable successful contemporary learning (Goodyear & Ret al.,is, 2010).

Research suggests that constructivist learning environments and learning designs are necessary to support pupils in becoming self-regulated learners (Dignath & Veenman, 2021; Hmelo-Silver et al., 2007). More specifically, several pedagogical elements have been identified as promoting the regulation of learning and should therefore be considered when designing phenomenon-based study modules. Such findings are largely based on classroom observations and teachers' self-reports (Adler et al., 2025; Dignath & Veenman, 2021; Ewijk et al., 2013; Perry & VandeKamp, 2000; Vosniadou et al., 2024).

The first important element of a constructive learning environment is to activate pupils' prior knowledge to help them understand learning goals and task requirements. Constructing goals in this way is important in collaborative processes for self- and co-regulation (Lim & Lim, 2020). Revisiting prior knowledge also encourages learners to monitor their own progress. To motivate active participation in learning, pupils should be provided with complex and open-ended problems and activities (Perry, 1998). Collaborative learning can also facilitate SRL, especially in complex tasks, as pupils can share the cognitive load (Dillenbourg, 1999). In addition, learning tasks should be integrated into real-life contexts to ensure the transfer of knowledge and to make learning meaningful for pupils (Perry, 1998). Accordingly, student-centered learning environments, where pupils can make choices about certain aspects of their learning—including content, setting, the level of challenge, and one's collaborative partners—have also been shown to promote SRL (Perry, 1998; Perry & VandeKamp, 2000). However, maintaining an appropriate balance between self-directed and teacher-guided learning, as well as providing timely teacher support, remains essential (Dignath & Veenman, 2021).

To function effectively as a pedagogical framework, phenomenon-based learning requires a structured and intentional learning design. Teachers must plan learning designs to operationalize pedagogical principles and support pupils' learning processes while exploiting the opportunities provided by technology (see, e.g., Conole, 2013; Mor & Craft, 2012). In technology-enhanced learning processes, learning design is generally created for an LMS as a structured sequence of materials, such as texts or videos, and assignments. During implementation, learning design mediates learning by shaping pupils' actions and guiding them toward the use of SRL strategies (Fan et al., 2021). A strong connection has also been identified between learning design and LA (Mangaroska & Giannakos, 2019), as learning design provides the theoretical framework for analyzing LA data (Mangaroska & Giannakos, 2019). To fully benefit from LMS learning designs, however, pupils require sufficient skills to regulate their own learning processes.

Scaffolding the regulation of learning in collaborative processes

In addition to pedagogical elements that promote SRL, primary students often require external support, such as scaffolding in the form of guidance or feedback, due to differences in their regulatory skills. Given that regulation becomes even more demanding in collaborative settings (Panadero & Järvelä, 2015), pupils typically require greater scaffolding assistance during collaboration (Järvelä et al., 2015; van Leeuwen & Janssen, 2019). Scaffolding refers to providing temporary support for learners to accomplish tasks they would not otherwise be able to complete by themselves (Wood et al., 1976). In blended learning environments, scaffolding for collaborative regulation can take multiple forms. Support may be provided by teachers, peers, digital platforms, or technological tools, and scaffolding can be either fixed or adaptive, as well as direct or indirect. Combining multiple scaffolding approaches may be particularly beneficial in primary education due to learners' diverse needs. As many pupils may still require direct instruction and teacher guidance to develop regulation skills (Dignath & Büttner, 2018), indirect scaffolding—such as that provided through the digital learning environment—can instead encourage learners to take control of their learning, actively engage in reflection, and adopt effective learning strategies.

Learning activity design itself can provide opportunities for indirect scaffolding in collaborative contexts. Learning design can offer cognitive scaffolding by organizing content, tasks, and assignments in ways that support collaborative regulation and reduce cognitive load associated with complex group work. In addition, designs incorporating prompts that encourage learners to articulate, reflect upon, and regulate their processes can function similarly to human tutoring support (Martin et al., 2019). Notably, these metacognitive scaffolds are considered central to indirectly supporting regulation (Järvelä et al., 2021) as they promote planning, monitoring, and reflection during collaboration (Molenaar et al., 2011).

Research also indicates that in digital learning environments scaffolding can enhance collaborative interaction, support shared goal formation through cognitive scaffolds, and assist groups in monitoring and regulating collaboration (Vogel et al., 2017). According to a meta-analysis of problem-based collaborative learning (Kim et al., 2020), digital metacognitive scaffolding and collaboration guidance effectively support academic achievements (Prasse et al., 2024; Shao et al., 2023). However, challenges remain in implementing technological scaffolding for collaboration. Students may deviate from designed learning pathways or ignore system recommendations, preferring to make independent decisions about their learning (Harley et al., 2018). Consequently, especially in the context of AI in education, it is anticipated that systems that excessively scaffold or over-regulate students' actions may hinder their ability to self-regulate their learning (Molenaar, 2022). Given that the objective of scaffolding is to gradually diminish support as students attain greater competency, it is imperative that scaffolding be adapted according to learners' specific needs (Azevedo et al., 2008; Molenaar, 2022).

LA visualizations offer further opportunities to scaffold collaborative processes by enabling students to reflect on their learning and assisting teachers in making data-informed pedagogical decisions. Given that the regulation of learning is understood as an ongoing sequence of learner actions and events (Molenaar & Järvelä, 2014), pupils' interactions within an LMS—such as accessing tasks, monitoring progress, or engaging in reflection—can serve as indicators of regulatory activity supported by LMS scaffolds. Temporal and sequential analysis of such data can reveal pupils' needs across different phases of collaborative learning. Thus, LA provides meaningful insights into pupils' behavior during collaborative processes (Wise et al., 2023). The possibilities arising from the utilization of AI in education make the use of this data even more beneficial, for example, through tutoring chatbots and adaptive learning technologies (Molenaar, 2022).

Moreover, insights gained through LA can assist educators in evaluating the real-time implementation of learning designs and making interventions (Ifenthaler et al., 2018) or improve pedagogical designs by identifying specific learning phenomena, including challenges or misconceptions (Mangaroska & Giannakos, 2019). Research has been conducted to examine individual learners' SRL through their digital traces, such as LA log data (Liu et al., 2024) and collaborative efforts within groups via epistemic networks (cf. Elmoazen et al., 2022). Nonetheless, a significant proportion of these studies overlook the implications and effects of pedagogical learning designs aimed at fostering SRL and predominantly focus on individual online learning processes within older demographic groups (Heikkinen et al., 2023).

Aims and research questions

As outlined above, there is still limited empirical knowledge on how pupils, especially those in primary education, collaboratively regulate their learning and use LMS as scaffolding resources (Dignath & Veenman, 2021; Prasse et al., 2024). This study addresses this gap by examining how an LMS-based learning design can *indirectly* scaffold pupils' regulation of collaboration during a collaborative study module. In addition, as research on the potential of LA to support blended collaborative learning processes remains scarce (Heikkinen et al.,, 2023), the study explores how LA visualizations may further support these regulatory processes.

To evaluate the implementation of the learning design, evidence was collected from both the physical environments (i.e., classroom observations) and digital contexts (i.e., LA log data) to gain a holistic and temporal understanding of the study module. First, the critical points and challenges encountered during the implementation of the learning design were examined through classroom observations. This was followed by an analysis of LA data, which reflects how pupils utilized the LMS to regulate collaboration during different phases of the process.

The specific research questions were as follows:

1) What are the critical phases and challenges encountered during the implementation of a collaborative phenomenon-based learning design that supports the regulation of collaboration in primary school?
2) How do pupil groups and individual pupils utilize learning design scaffolds to support SRL and the regulation of collaboration during different phases of the process?

Methods

The pedagogical design

For this study, a specific study module was designed and implemented within the LMS (see Table 1). The learning design of the study module was formulated to promote SRL and SSRL in two ways: (1) through the pedagogical learning design and (2) by providing both cognitive and metacognitive scaffolding support at different phases of the process. Elements previously identified as critical for successful collaboration (see, e.g., Dillenbourg, 1999) guided both the design and implementation phases.

The pedagogical design of the study module was based on phenomenon-based learning, which is emphasized in the Finnish national core curriculum for basic education (Finnish National Board of Education, 2014). Accordingly, pedagogical elements known to promote the regulation of learning (see Sect. "Phenomenon-based learning design to promote the regulation of learning") were central in the design. Therefore, the planned study module started by scanning pupils' pre-knowledge, then emphasizing pupils' collaborative knowledge-building, before pupils worked together as active producers of media artifacts. The learning design was highly student-centered and open-ended, highlighting opportunities offered by digital environments and allowing pupil groups to make decisions concerning their own learning processes.

Table 1 The learning design of the study module Sustainable Development

	Sustainable Development **Duration—2 weeks** **Group learning assignment: Create a media artifact about the selected product life cycle**	
Lesson (90 min)	**Subject**	**Contents of the lesson**
1st	Start of the study module	- Prior knowledge of sustainable development - Understanding the foundational concepts of sustainable development - Overview of the study module, including general objectives and group assignment - Pre-knowledge of the chosen product of the group
2nd	Start of group work: collaborative knowledge construction	- Sharing prior knowledge within groups - Learning how to search for reliable information resources - Searching for information about the life cycle of the chosen product
3rd	Manuscript	- Finalizing the information search - Choosing the media format of the artifact - Writing the manuscript - Returning the manuscript
4th	Producing the artifact	- Start making the group artifact - Learning different camera techniques - Using a checklist for artifact production
5th	Producing continues	- Learning about voice recording - Using a checklist for artifact production
6th	Editing and formatting	- Learning to edit a video - Returning the artifact
7th	Show time	- Presenting each group's artifact - Learning to deliver a presentation - Providing feedback on other groups' artifacts
8th	Final assignment	- Final learning assignment: individual reflective writing

The subject of the study module was sustainable development. Its primary objective was for pupils to understand the basic principles of sustainable development, such as the sustainable use of natural resources and human impacts on the environment. These objectives were addressed by investigating the lifecycle of an everyday object. Broader goals of the module also focused on developing competencies such as group work skills, collaboration, and information-seeking abilities. Emphasizing competencies has been shown to support the development of self-regulation (Schunk & Swartz, 1993). The central assignment required pupils to investigate which materials, industrial processes, and transportation stages are involved in producing an everyday object and how the object could be recycled after use. All tasks were designed as open-ended activities, allowing pupils to make their own choices and set group-specific goals, such as selecting the object of investigation and choosing the format of the digital learning artifact. Such autonomy supports motivation, which in turn promotes SRL (Schunk & Zimmerman, 2008).

The learning design was developed in close collaboration with teachers from a university teacher training school and researchers. The participating teachers were experienced in implementing phenomenon-based learning, while the researchers were responsible for designing the LMS environment to support pupils' learning processes and SRL. Pre-service teachers completing their teaching practice in the participating

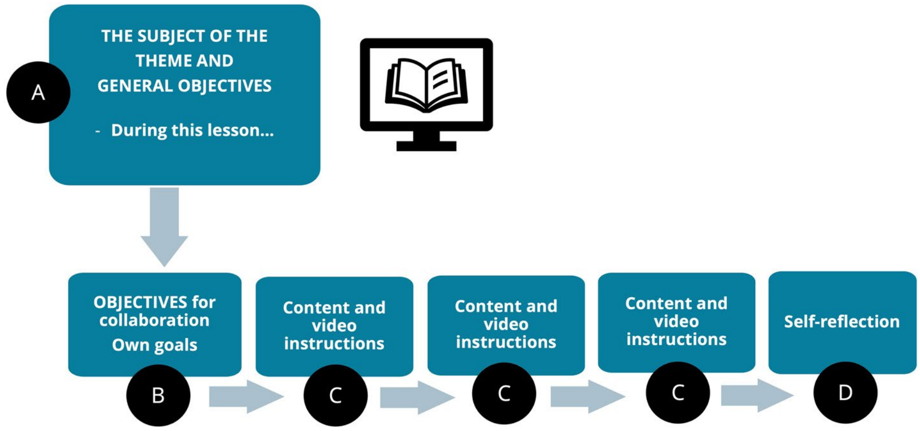

Fig. 2 The general structure of the design of a study module lesson supporting SRL with objectives (**A** and **B**), content (**C**) and reflection (**D**)

classrooms contributed to producing materials used in the LMS and were also responsible for conducting some lessons.

The design for scaffolding the regulation of learning

Socially shared planning, monitoring, and reflection do not occur spontaneously; rather, they need to be intentionally supported through instructional design. Therefore, the design of the study module LMS was intended to promote pupils' regulation of learning (Panadero, 2017; Zimmerman, 2002). This was achieved by structuring the materials to support the different phases of SRL to make the process cyclical (see theory 2.1). The general design of a theme, repeated in each of the eight lessons, is presented in Fig. 2. The LMS materials guided pupils to set goals, monitor their group process and collaboration, and ultimately reflect on their learning within each theme.

Providing opportunities for self-evaluation is an effective way to indirectly activate the learning regulation (Perry & VandeKamp, 2000), as such activities prompt pupils to monitor their learning. Establishing a coherent structure for metacognitive monitoring and reflection (cf. Fleck & Fitzpatrick, 2010) is therefore particularly important for scaffolding pupils' SRL. Accordingly, metacognitive tasks were embedded in the LMS as an essential part of the learning design and were introduced through reflective questions at the end of each lesson. Pupils responded to these questions individually, while the objectives for reflection were introduced at the beginning of each lesson, with particular emphasis on collaboration skills.

To scaffold the cognitive load of collaborative learning, the study module was divided into eight lessons, each with its own subject and objectives to guide the process (see Table 1 and Fig. 2, point A). Each lesson in the LMS followed the same basic design: it began by introducing the lesson goals (Fig. 2, point B), followed by content and instructions designed to guide and support the group process (Fig. 2, point C), and concluded with a self-reflection activity aligned with the objectives of the theme (Fig. 2, point D).

Research context and participants

The study module was implemented in two classes at a Finnish university teacher training school. Participants included 82 pupils: 38 fifth graders and 44 sixth graders, aged 11–13, all accustomed to actively using tablets in their learning. Learning in both classes was organized in large groups supported by two classroom teachers, a shared special education needs teacher, teacher assistants, and pre-service teachers completing their teaching practice. The sixth-grade pupils had more prior experience with pedagogically similar learning processes than the fifth graders. During lessons, both classes worked in a semi-open, flexible learning hub where each class had its own designated classroom area. The study module was conducted over two weeks in spring 2021 and consisted of eight lessons, each lasting 90 min.

During the module, pupils worked both face-to-face in small collaborative groups of two to four pupils and within the LMS using their tablet devices. The LMS-based learning design guided both teachers' and pupils' actions throughout the learning process. At the beginning of each lesson, the teachers introduced lesson goals and materials, while at the end pupils were guided to complete individual reflections. However, in the first and last lessons, learning was more teacher-centered due to the phases of the module.

Collaboration in groups was encouraged by highlighting the social skills needed for collaboration in the objectives of the entire module and lessons. Moreover, pupils were encouraged to share roles and responsibilities within their groups. Both pupils and teachers were able to follow progress through simple LA dashboard visualizations displaying progress as percentages and star indicators (cf. Väisänen et al., 2022). Pupils were free to choose the format for their final digital artifact describing the lifecycle of their chosen product. The module concluded with documentary festival events, during which each group presented its artifact, such as a video, a Minecraft world, or a slide presentation.

The pupils and their parents were informed about the nature of the research through briefings and an explanatory video. All collected data were anonymized prior to analysis. Ethical approval for the study was granted by the University of Eastern Finland Institutional Review Board for Research Ethics (11/2020), and national ethical guidelines for research involving human participants (Kohonen, Kuula-Luumi, & Spoof, 2019) were followed throughout the study.

Data collection and analysis

Mixed methods (Creswell & Plano Clark, Creswell & Plano Clark, 2018) were used to collect and analyze data in order to gain a holistic and multifaceted understanding of the blended learning process. Combining qualitative observation data with quantitative LMS log data helped address methodological gaps identified in both SRL research (Gambo et al., 2021) and LA research (Tempelaar et al., 2021) that call for the integration of multiple data sources.

Both the classroom observation data and LMS log data were collected during the implementation of the study module. First, two researchers observed each lesson (16×90 min), excluding the first author, who worked as a classroom teacher in one of the participating classes. Classroom observation was used to analyze the collaborative face-to-face activity in the classroom (Eradze et al., 2019). The chronological actions of both pupils and teachers were recorded as time-stamped field notes. Second, the LMS automatically collected log data on pupils' behavioral actions during the lessons, such as accessing content materials, viewing instructions, or submitting assignments.

Qualitative content analysis

The classroom observation resulted in 48 pages of written text. To address the first research question, the observation data were analyzed using qualitative content analysis to identify pedagogical challenges emerging during the collaborative study module. According to Hsieh and Shannon (2005), the goal of content analysis is to develop an understanding of the phenomenon under study through a systematic process of coding and identifying themes or patterns. In this study, the data were first read through several times to ensure familiarity with the content. Coding was conducted using Atlas.ti (version 24). An inductive, data-driven content analysis approach was applied (Elo & Kyngäs, 2008; Graneheim et al., 2017). In the first phase, open coding enabled insights to emerge directly from the data, generating context-specific insights for the participating classrooms and the study module. For example, situations in which pupils expressed a need for additional support were coded as challenges in understanding the assignment. After the initial coding, the codes were reviewed to identify similarities and broader themes (Elo & Kyngäs, 2008), with similar codes being merged into broader categories. The analysis proceeded iteratively through multiple rounds of code and data review to ensure the categories accurately reflected the observations. The final categorization identified eight distinct types of challenges. Temporal aspects were also examined to determine at which phases of the study module specific challenges most frequently occurred. Initial coding was conducted by the first author, after which the third author reviewed and discussed the categories, leading to minor adjustments to validate the findings.

Sequence analysis

To address the second research question, LMS log data automatically collected during the lessons were analyzed to examine pupils' actions within the LMS. In total, we collected 2589 logs (1413 for fifth-graders and 1176 for sixth-graders). The log data were

Table 2 Codes created for sequence analysis

Code	Description
Assignment	Special learning assignments requiring additional work from pupils
Content	Content material for learning, including texts and visuals
ExtraContent	Supplementary material for fast and interested learners
GroupAssignment	Learning assignments requiring group work
Instruction	General guiding materials
MCTask	Metacognitive tasks for self-assessing and reflecting on the learning
Objectives	Objectives for each lesson
Presentation	Learning artifacts made by pupils
Scaffold	Mandatory materials to help pupils to start knowledge building
Task	Smaller, multiple-choice tasks
VideoContent	Video material for learning the content

organized according to the lesson timetables, and actions were coded to describe how pupils used the LMS features to support their regulation of learning in different phases of the module (see Table 2). Sequence analysis was then conducted to explore how pupils used the study module LMS to regulate their learning in different phases of collaborative learning (following Saqr et al., 2024). The R package *TraMineR* (Gabadinho et al., 2011) was used to build a sequence object from students' LMS logs. Unlike many sequence analysis applications focusing solely on clickstream order, this analysis accounted for both the sequence and duration of actions. Following approaches used in previous studies (López-Pernas et al., 2022; Paavilainen et al., 2024), SPELL sequences were employed, in which each LMS interaction had a clear beginning—the instant of the click—and an end—the instant in which the subsequent click took place. This approach made it possible to estimate the time pupils spent interacting with each LMS resource. In the resulting data, each step in a sequence represented one minute of activity during intervention sessions. Index plots were used to visualize sequences by collaborative group (groups 1–26), session (sessions 1–7), and grade level (grades 5 and 6), enabling comparison of how groups' LMS use evolved over time.

Lastly, the results of the qualitative and sequence analyses were integrated by comparing the challenges identified through observation against the log data sequences individually corresponding to each lesson. These comparisons were then examined in relation to the LMS learning design to clarify the contextual conditions and specific phases of the learning process in which the challenges emerged.

Results

In this section, we begin by presenting the results of the qualitative content analysis of classroom observations and log-data sequence analysis. The Discussion section then integrates these findings, with particular attention to pedagogically smooth and challenging phases of the learning process.

The results of the observation: pedagogical challenges of collaborative learning

Based on a qualitative content analysis of the observation data, several pedagogical challenges were identified during the collaborative learning process. Temporal analysis identified two particularly challenging phases: the beginning of group work and the finalization of the media artifact prior to the deadline (see Fig. 3). In contrast, collaboration proceeded more smoothly during the middle phase of the module, particularly in lessons four and five, when all groups were actively engaged in developing their media artifacts. Similarly, the first and last lessons were smoother as these phases were heavily teacher-guided. It is important to note that the processes varied between the two groups: overall, sixth-graders encountered fewer challenges than fifth-graders during the lessons.

Understanding and staying on task Certain groups struggled to remain focused on the learning assignment. Instead, they engaged in unrelated discussions or browsed the Internet. As a result, certain pupils had difficulty understanding the assignment and its objectives. This issue was particularly prevalent in the fifth-grade class, especially during the initial three lessons. However, some groups, particularly those in the sixth grade, were able to efficiently regulate their learning and quickly proceed with their assignments, as the following quotation from the classroom observation notes demonstrates:

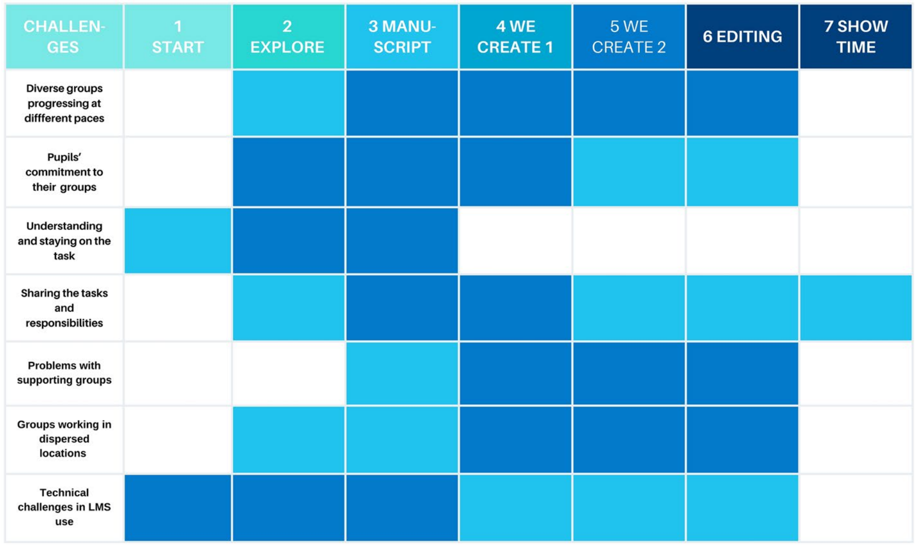

Fig. 3 Challenges across different phases of the study module. The intensity of the color shading represents the degree of challenge encountered, with darker shades indicating greater challenges

> *After 45 minutes, some groups were well into planning and brainstorming, while in other groups, some pupils still had uncertainties about what was expected of them or had difficulties concentrating on the task. (Class 5, 3rd lesson)*

Pupils' commitment to their groups At the beginning of the learning process, groups were formed according to pupils' choices and ultimately by teachers' decisions. Some pupils struggled to integrate into their assigned groups, leading to disruptive behaviors such as moving between groups or avoiding responsibility. Pupils unwilling or unable to participate demanded considerable teacher attention. This challenge was observed mainly among fifth graders and persisted throughout the module:

> *The wandering has begun, and the pupil visits the group in front. Teacher 4 commands him to return to his place, but the pupil continues to Jari's group. Teacher 4 instructs him to return, and finally, he does. (Class 5, 4th Lesson)*

Diverse groups progressing at different paces Groups differed in their collaboration skills, pace of progress, and need for scaffolding. While some groups—particularly in sixth grade—were eager to move forward, others struggled to continue their work. As a result, groups progressed at different speeds. Many required additional assistance, such as technical support or help organizing workspaces, especially toward the end of the module:

> *Some pupils are wandering around the classroom, and some haven't started working even after 45 minutes. Some groups have worked very diligently and have independently produced a script. Some groups need a lot of guidance and direction in their work. (Class 5, Lesson 3)*
> *Someone has already asked if they can continue where they left off yesterday, but Teacher 2 says that first, certain things will be discussed before continuing with the work ... The script is still unfinished for three groups, so they need to finalize it. Oth-*

erwise, the work will continue in the lesson, and help will be asked if necessary. A pupil points out, asking what if the work is already finished. (Class 6, 5th Lesson)

Sharing the tasks and responsibilities Teachers encouraged groups to distribute responsibilities and adopt different roles within teams. Nevertheless, in some groups, workloads were not shared equally, as illustrated below:

One pupil is wandering around the classroom and not working with their group. In one group, a pupil asked who would help them, but the other two group members just stared at their tablets. (Class 5, 6th Lesson)

Groups working in dispersed locations across the learning environment During artifact production—particularly in lessons four to six—groups moved to different parts of the learning environment or even elsewhere in the school to record videos or podcasts. This dispersion made it difficult for teachers to monitor and support all groups effectively:

Jenni's group has also disappeared from the classroom. Teemu's group has moved to the hub´s lobby and is preparing for filming, with Teacher 3 helping to adjust the tripod. The news girls are filming at the back of the sixth-grade classroom. Timo's group is in the soundproof room. (Class 5, 4th lesson)

Problems with supporting groups During the more challenging phases of the learning process, particularly towards the conclusion of the study module, teachers did not have sufficient time to address each group's needs. As a result, some groups were left without the support they needed. In addition, some pupils hesitated to ask for help even when needed, and some struggled to make use of the support offered.

Technical challenges in LMS use Although pupils were familiar with the LMS, some still required assistance, particularly when new LMS functions were introduced. Teachers, especially in fifth grade, responded by providing more structured guidance in LMS use, particularly at the beginning of the module.

The results of the LMS log data sequence analysis

Based on the LMS log data, sequence analysis was used to visualize how pupils utilized the LMS to regulate their learning in different phases of the study module. In the index plots, each horizontal bar represents an individual pupil's sequence of actions. Colored segments indicate different types of LMS actions, and the length of each segment reflects the time spent on that activity. Sequences are presented by collaborative group, with separate figures for each class (see Appendices 1 and 2).

Generally, the sequence analysis revealed meaningful variation in LMS use across classes and lessons, despite the LMS design remaining largely consistent throughout the module. This variation highlights the impact of teachers' pedagogical implementation (cf. Paavilainen et al., 2024). In the following paragraphs, we focus on selected lessons based on the findings of the qualitative phase: the beginning of the process (1st and 2nd lesson) and the finalization phase of the group project (6th lesson). In addition, specific observations from 4th lesson are presented. Visualizations of the entire process are provided in Appendices 1 and 2.

At the beginning of the process (1st and 2nd lesson, Fig. 4), pupils utilized the LMS, particularly in the fifth-grade group, as these lessons were more teacher-centered and

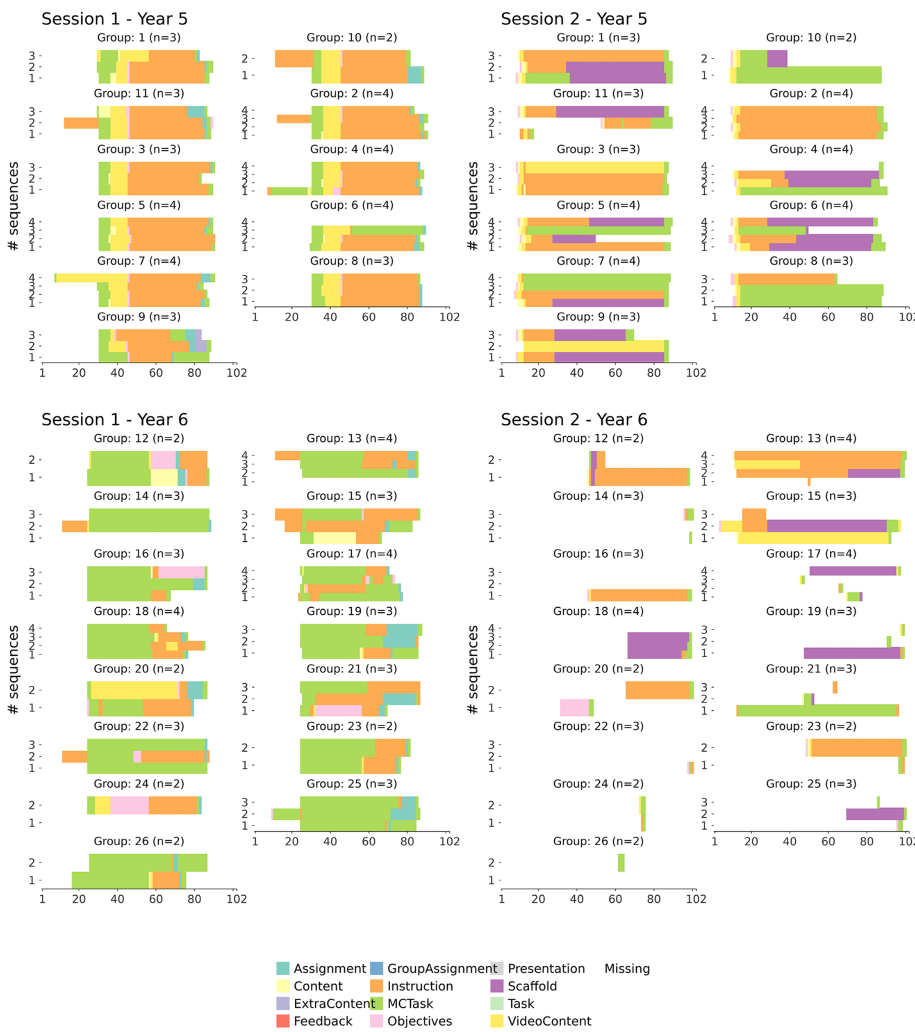

Fig. 4 The sequences visualizing the lessons 1 and 2 (sessions), class groups (years) 5 and 6

pupils followed the instructions given by the teacher. Sequences are consistent within a group and across different groups. Consequently, action sequences appear relatively consistent both within and across groups. However, the log data indicate that pupils rarely accessed the stated lesson objectives. Instead, they focused primarily on task instructions and supporting materials needed to initiate their group assignments. As collaboration began in 2nd lesson, differences between class groups became more pronounced. Fifth-grade pupils continued to engage with the LMS in a systematic and comprehensive manner. In contrast, sixth-grade pupils used the LMS more selectively, accessing materials only when necessary. In several sixth-grade groups, only some members consulted the instructions or scaffolding materials provided in the LMS, suggesting a more distributed or strategic use of digital scaffolds within groups.

Later in the process, pupils no longer utilized the LMS to regulate their learning. The results from lesson 6 (Fig. 5) indicate that pupils primarily accessed the LMS to complete the individual metacognitive reflection task at the end of the lesson, often doing so briefly. In particular, fifth-graders largely discontinued active LMS use after Lesson 4.

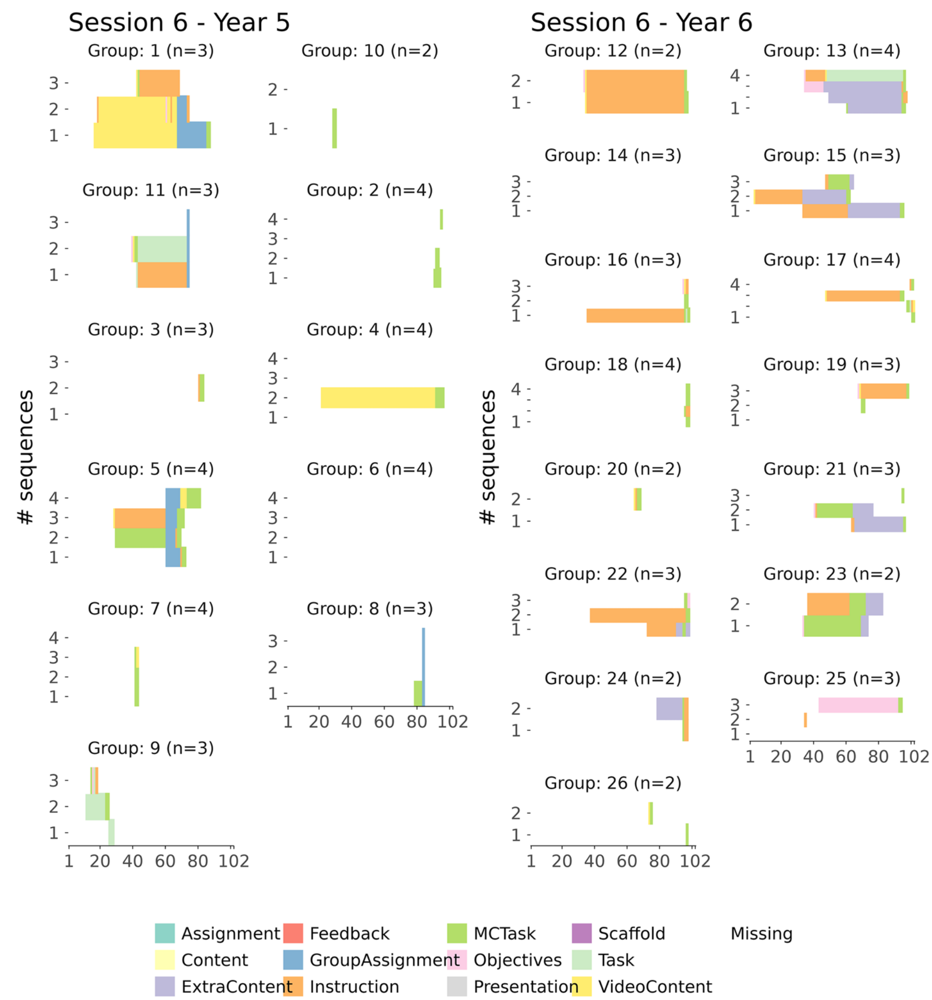

Fig. 5 The sequences visualizing lesson 6 (session), class groups (years) 5 and 6

In contrast, although sixth graders began using the LMS more selectively from Lesson 2 onward, their engagement appeared to increase again toward the end of the module. By Lesson 6, several pupils revisited instructions and accessed additional content beyond the mandatory reflection task. This suggests a more purposeful or need-based use of the LMS in the finalization phase.

As a result, the sequences became more differentiated not only between groups but also within individual groups—both in terms of the time allocated and the actions performed in the LMS. One exception to this differentiation was observed during the group assignment in the fourth lesson (and in the sixth-grade class also in Lesson 5; Fig. 6), where groups were required to evaluate the progression of their learning process using a checklist. The log data analysis underscores the significance of this specific assignment, particularly within the fifth-grade group, as many pupils opened the assignment and devoted substantial time to its completion.

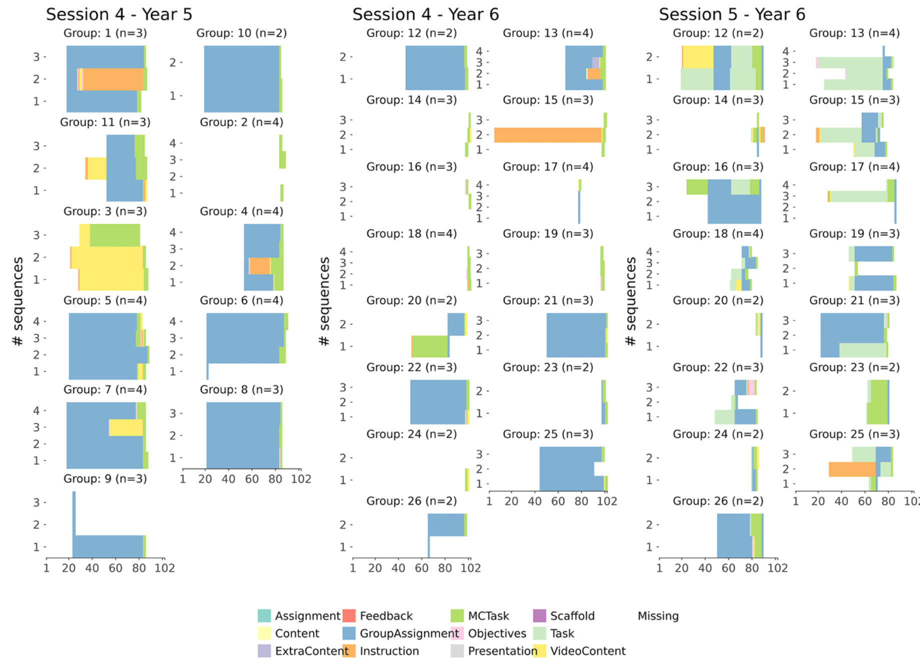

Fig. 6 The sequences visualizing lesson 4 in the fifth grade and lessons 4 and 5 in the sixth grade

Discussion

This study explored how an LMS-based learning design that indirectly scaffolded SRL functioned as regulation support for groups engaged in collaborative, phenomenon-based learning in a blended primary school context. First, critical phases of the collaborative, open-ended learning process were identified through classroom observation. Next, LA log data were analyzed to explore how pupils used the LMS to regulate their learning. In this section, the results of the classroom observations and LA log data are merged to form a holistic, combined understanding. This is followed by a discussion in which the findings are examined in light of prior research.

Beginning of the group work—technical challenges and understanding the assignment According to classroom observations, the second lesson emerged as the most pedagogically challenging phase. During this session, groups were formed and collaborative work officially began. Pupils needed to adjust socially to their new groups, understand the assignment, and manage technical aspects of working in the LMS. These demands led to behavioral challenges, technical difficulties, and confusion about expectations. As the log data for Lesson 2 show, fifth-graders utilized the LMS to address these challenges. They revisited the instructions, consulted the scaffolding materials, watched video content, and completed metacognitive tasks. In contrast, sixth-graders demonstrated a more selective pattern of LMS use. Often, only certain group members accessed instructions or scaffolding materials, while others assumed different responsibilities. Most sixth-graders completed the individual metacognitive reflection tasks at some point during the lesson, but overall LMS engagement was less systematic. This may indicate a more distributed form of regulation within groups, but it also suggests that the LMS was not collectively leveraged as a shared regulatory resource.

Smooth(er) collaboration to produce the media artifact In Lesson 4, groups had already developed a shared vision for the manuscript and the media artifact. They collaborated to ensure that sufficient information was shared while simultaneously producing different media formats. Log data revealed one particularly notable feature: intensive use of a group checklist assignment. This checklist required groups to evaluate whether they were on schedule and whether required components were completed. To complete the task, group members needed to negotiate responsibilities, review individual contributions, and collectively mark progress. In addition to supporting groups' own monitoring, the checklist provided teachers with insight into group progress.

During this phase, apart from the checklist, LMS usage for regulatory purposes decreased. This suggests that once collaboration stabilized and shared understanding was established, pupils relied more on face-to-face interaction than on digital scaffolding. Because media production largely occurred outside the LMS environment, pupils may have perceived less need to consult it.

As the phenomenon-based study module was implemented across two diverse classrooms, the analysis was able to provide insights into implementation variability and help us understand which results stemmed from the module itself and which were influenced by classroom-specific factors. Accordingly, this comparison also revealed interesting differences between classes. Generally, fifth-graders faced more challenges during the process. This may relate to less experience with phenomenon-based pedagogy or less developed regulatory skills. LMS use in the fifth-grade class also appeared more teacher-driven, aligning with previous findings (see also Paavilainen et al., 2024).

Furthermore, the openness of phenomenon-based learning caused pedagogical challenges for pupils in regulating their learning and, consequently, for teachers looking to facilitate this process. According to Järvenoja et al. (2019), collaborative learning often involves task-related and socio-emotional challenges. In this study, difficulties included understanding the assignment, maintaining focus, distributing responsibilities, managing diverse timetables, and navigating technical aspects of the LMS. Although challenges occurred throughout the process, both observation and log data indicate that the beginning of the collaboration was particularly critical.

At the same time, the log data indicate that pupils did not consistently utilize the LMS to scaffold their learning throughout the collaborative blended learning process (see also Harley et al., 2018). During the challenging initial phase, groups used the instructions or materials in the LMS, which can be interpreted as an attempt to create shared task understanding (Malmberg et al., 2022). Given the crucial role of the goal-setting and planning phase for the success of the collaborative process (see, e.g., Miller & Hadwin, 2024), this time allocation is highly important. However, as learning progressed, the role of the LMS diminished. This is inconsistent with the notion that, according to previous research, pupils require support for their regulation of learning in each phase of the learning process (Baars & Viberg, 2022). One explanation for this may be that group-level regulation reduced the need to consult the LMS. Additionally, as the creation of the media artifact mostly took place outside the LMS in face-to-face interaction, pupils may have overlooked its potential for supporting regulation. These findings are supported by Järvelä et al., (2016), who highlight the need to design prompts for goal-setting, planning, and motivation regulation to support socially shared regulation of learning.

This study thus reinforces the idea that regulation in collaborative processes is dynamic and temporal, requiring different scaffolding strategies at different phases. While the LMS design followed similar principles across themes, pupils may have felt that it did not fully respond to their evolving needs, which may explain the limited use of LMS-based scaffolding. However, research also shows that significant benefits can be achieved when leveraging technology to support the regulation of learning (Prasse, 2024). Therefore, future designs should more carefully consider the temporal progression of collaborative processes. Accordingly, following Sharma et al., (2024), the possibilities offered by adaptive LMS systems will presumably constitute one potential solution for scaffolding the regulation of collaboration in more student-centered learning processes in the future.

In addition to the phases of the collaborative process, the comparison between the LMS design and log data shows that the learning assignments provided in the LMS also influenced how pupils engaged with the LMS to regulate their learning. More precisely, this study demonstrates the need for functional group monitoring tools to enhance collaborative learning. As Järvelä et al., (2015) argue, learning environments should scaffold group regulatory actions and prompt reflection on shared strategies. The checklist assignment provided metacognitive scaffolding and appeared to help pupils share monitoring responsibilities and maintain focus on task completion, as previously noted by Malmberg et al., (2015). Compared to individual metacognitive reflection tasks at the end of lessons, substantially more time was devoted to the checklist assignment, highlighting its importance. This finding aligns with Shao et al., (2023), suggesting that the most effective scaffolding for SRL and SSRL is achieved through the utilization of group awareness tools. The checklist assignment also functioned as a summarizing mechanism, enabling groups to negotiate shared goals and required actions. The value of summarizing elements in collaborative knowledge construction is supported by Wise and Chiu (2011). Similar metacognitive monitoring tools could be implemented in other phases of the process to enhance learning. Given findings by Baars et al., (2018) that metacognitive monitoring often lacks accuracy during challenging phases, structured monitoring support may be especially important at the beginning of collaboration.

Overall, this study highlights the significant correlation between learning design and LA (cf. Ifenthaler et al., 2018; Mangaroska & Giannakos, 2019; Persico & Pozzi, 2015). Log data analysis provided insights into how pupils used the LMS to regulate collaboration and how the LMS design could be further developed to scaffold challenging phases. To fully leverage LA for informing pupils about their collaborative contributions and supporting data-informed teacher interventions, LMS design must be carefully aligned with pedagogical intentions. Firstly, the pedagogical design of the LMS must comprehend and adapt to the temporal progression of the collaborative process. Secondly, the resources or assignments provided within the LMS must engage students and groups to effectively utilize the platform. In this study, indirect scaffolding alone was not sufficient to sustain LMS use. Future designs could therefore integrate more dynamic LA feedback and visualizations to provide engaging scaffolding. AI-based tools, such as tutoring chatbots, may also offer new possibilities for delivering feedback and promoting SRL in more interactive ways. Thus, the insights gained in this study are relevant not only for LMS design but also for the development of educational systems integrating LA and AI.

Conclusions

This study contributes to research on supporting primary pupils' collaborative processes and SSRL. The findings have implications for researchers, learning designers, and teachers. First, the findings show that the regulation of learning in collaborative projects is dynamic and phase-specific and therefore requires more adaptive and phase-aware scaffolding. LMS-based scaffolding appears to be especially important at the beginning of collaborative and open-ended learning processes. In addition, summarizing group awareness tools, such as the checklist assignment, provides opportunities for shared metacognitive monitoring and thus contributes to the regulation of collaboration.

Secondly, to utilize the full potential of LA for scaffolding collaborative processes, this study highlights the importance of aligning pedagogical principles of learning design with LA. To provide meaningful insights for teachers, LA should be designed to scaffold the phases and objectives of the learning design. Thus, through more powerful visualizations, teachers can receive feedback and make changes to their learning designs already during the process to maximize the potential of LMS to support collaboration (see also Adler et al., 2025; Ifenthaler et al., 2018).

However, in spite of these insights, it must be acknowledged that this study is not without limitations. As we prioritized conducting this study within a real classroom context in an authentic pedagogical and technological setting, its findings are limited to a small cohort of pupils and one specific learning design and limited timeframe. In future research, diverse designs, longitudinal effects, and a greater number of pupils would be needed to confirm and extend these findings.

Moreover, in a real-life setting, not all contextual factors can be fully controlled, which may influence the observed outcomes. In the context of this study, teachers' pedagogical actions may have affected pupils' LMS usage, as further discussed by Paavilainen et al., (2024). Moreover, because the advanced log data analysis methods used in this study were not available during the implementation phase, we were unable to examine how real-time visualizations might have influenced teachers' decision-making, interventions, or adaptations of the learning design. This should be investigated in future research.

Future research combining the aspects of the regulation of collaboration, learning design, and LA could investigate the phases of collaborative learning more thoroughly to understand the different forms of scaffolding needed in each phase. Additionally, as this study focused primarily on the pedagogical process, future research could investigate the effects of such learning designs on pupils' learning outcomes, including both content knowledge and skill development.

Appendix 1

See Fig. 7

Appendix 2

See Fig. 8

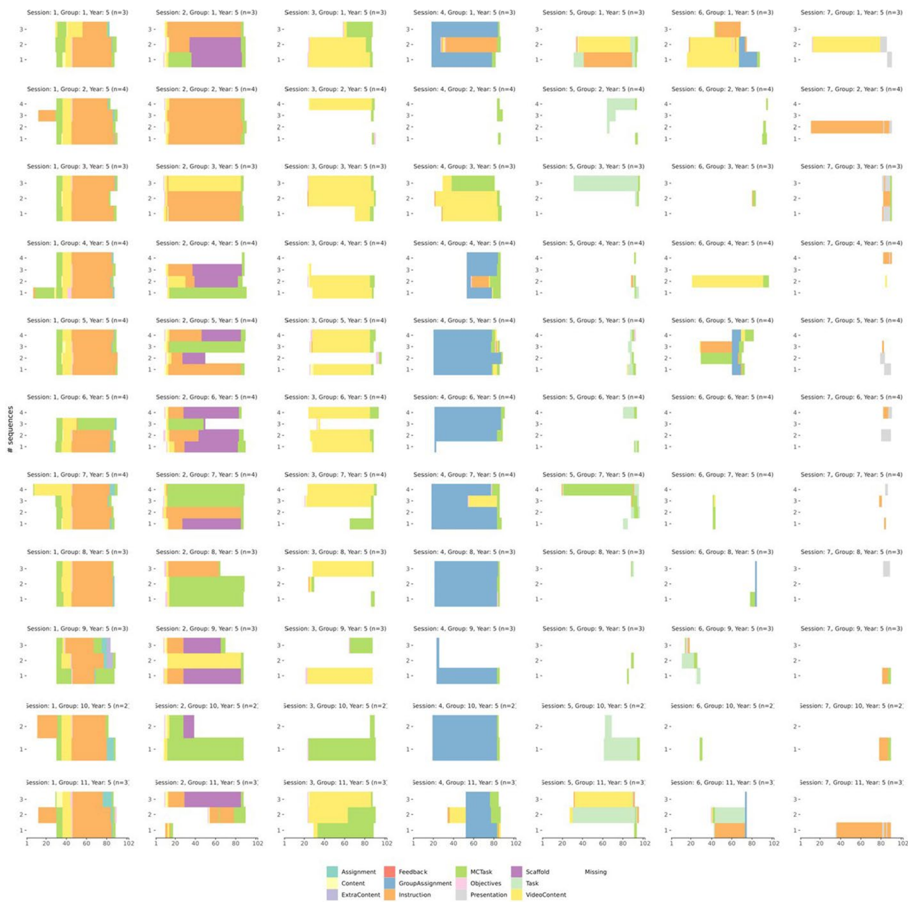

Fig. 7 Fifth-graders' index plots

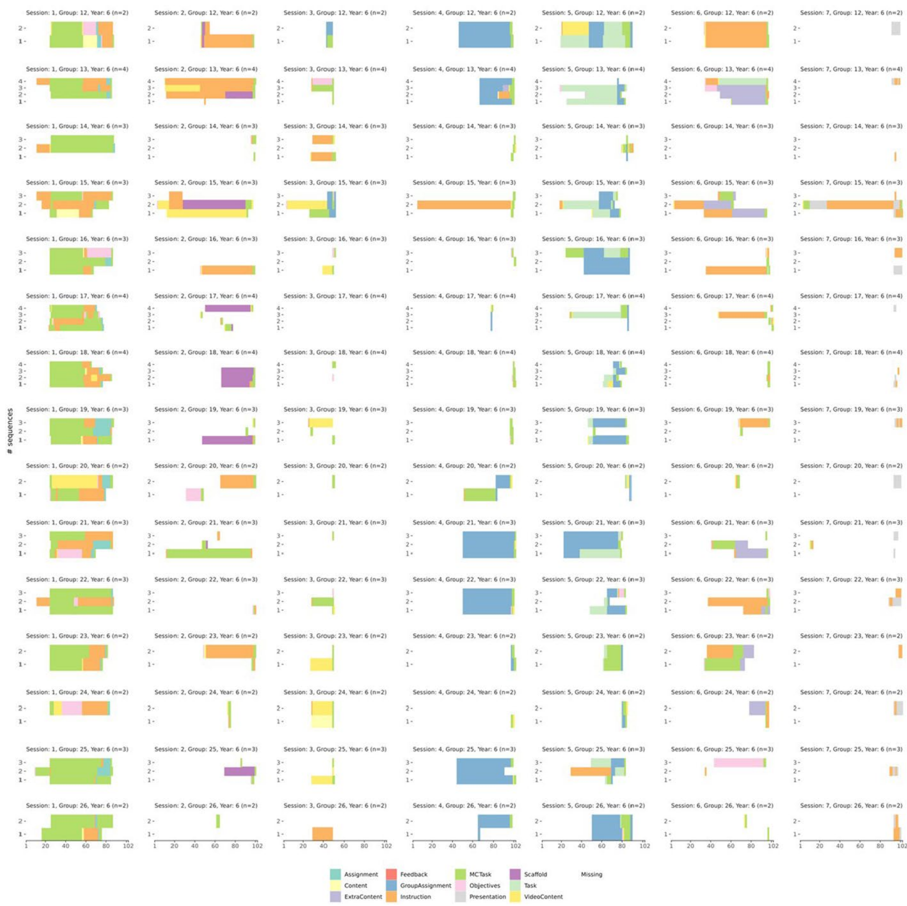

Fig. 8 Sixth-graders' index plots

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Author contributions
All authors played a role in the conception and design of the study. Sanna Väisänen and Laura Hirsto handled the material preparation and data collection. The initial analysis was conducted by Teija Paavilainen, Sonsoles López-Pernas, and Laura Hirsto. Visualizations were created by Sonsoles López-Pernas and Teija Paavilainen. The first draft of the manuscript was prepared by Teija Paavilainen, with input from all authors on earlier versions. All authors reviewed and approved the final manuscript.

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Data Availability
The datasets generated and analyzed during the current study are not publicly available due to privacy concerns.

Declarations

Competing interests
The authors have no competing interests to declare that are relevant to the content of this article. In designing and implementing the study, the ethical recommendations of the Finnish Advisory Board on Research Integrity were followed. The Committee for Research Ethics (University of Eastern Finland) approved the design of the study.

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